COMMUNITY ENGAGEMENT PROJECT (CEP) • 2026–2027

EcoQuest — Campus Restoration Game

An Interactive Mobile Application for Fostering Student Sustainability Habits

PRESENTER INFORMATION

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Program & Branch: B.Sc. Computer Science (TY Sem V)

Department: Department of Computer Science

INSTITUTION & PLATFORM SPECIFICATIONS

College / University: SMT. CHM COLLEGE, Ulhasnagar-3

Academic Year: 2026–2027

Faculty Guide: [Insert Faculty Mentor Name & Designation]

Platform: React Native 0.86.3 • Expo SDK 57 • Android

The Campus Sustainability Dilemma & Student Motivation

📢 Passive Communication

Campus guidelines rely on static posters, circulars, and notices that are largely ignored by digital-native students.

🌊 The 'Drop in the Ocean' Bias

Students often believe individual micro-actions (switching off one light or carrying a bottle) have no measurable impact.

📉 Absence of Immediate Feedback

Unlike academic grading or sports, sustainable daily habits offer zero milestone markers, levels, or social recognition.

💡 THE ECOQUEST POSITIVE REINFORCEMENT LOOP

• **Actionable Daily Quests:** Micro-challenges broken into simple, achievable campus habits (Energy, Waste, Water, Transport, Food).

• **Instant In-App Gratification:** Immediate positive reinforcement awarding verified Lifetime Points and Spendable Funds.

• **Tangible Campus Restoration:** Reinvesting points funds physical restoration projects (Shade Trees, Solar, Compost, Water Station).

• **Visible Visual Transformation:** The central quad evolves dynamically on-screen from a dormant yard into a thriving green sanctuary.

ECOQUEST CEP PRESENTATION

Project Objectives & Demonstration Scope

🎯 Daily Quests

8 micro-challenges across 5 sustainability domains awarding up to 210 pts.

⚖️ Dual Economy

Separates lifetime score from spendable funds, encouraging upgrades without penalty.

🌳 Visual Campus

Cumulative 5-stage native illustration reflecting physical green infrastructure.

🏆 Progression

4 player levels, 8 milestone badges, and an 8-peer dynamic leaderboard.

🔬 ACADEMIC SCOPE & DEMONSTRATION BOUNDARIES (CEP INTEGRITY)

- **Educational Demonstration Prototype:** Developed as an academic prototype modeling incentive structures rather than measuring physical carbon offsets.
- **In-App Self-Reported Actions:** Actions are recorded via in-app button presses; no IoT sensors, camera AI, or GPS tracking are required.
- **Offline Local Client:** Runs entirely on the phone via AsyncStorage. Zero cloud server costs, zero login barriers, and complete data privacy.

Technology Stack & Unidirectional Architecture

🛠️ VERIFIED TECHNOLOGY STACK

- **React Native 0.86.3:** Cross-platform mobile UI runtime with native Android layout primitives.
- **Expo SDK ~57.0.30:** Managed build tooling, Metro bundling, and live Android execution via Expo Go.
- **React 19.2.3:** Component architecture using useState, useEffect, useContext, and useRef.
- **AsyncStorage 2.2.0:** Offline key-value persistence under 'ecoquest.gameState.v1'.
- **Safe Area Insets ~5.7.0:** Automatic clearance for Android 3-button nav (48dp) and gesture bars.
- **Vanilla StyleSheet:** High performance, memory-efficient styles without bundle bloat.

📐 UNIDIRECTIONAL DATA FLOW & SELECTORS

1. Single Source of Truth: GameProvider context maintains completedQuestIds and purchasedUpgradeIds ground truth arrays.

2. Pure Selector Pattern: Lifetime Points, Spendable Fund, Stage, Level, and Badges computed on-the-fly to prevent stale state.

3. Storage Serialization Mutex: queuePersist promise queue guarantees rapid taps write sequentially to AsyncStorage without race conditions.

4. Startup Schema Sanitization: validateSavedGameState() validates loaded JSON, rejects corrupt payloads, and heals non-sequential upgrades.

Home Dashboard & Campus Restoration Concept

🏛️ CUMULATIVE 5-STAGE CAMPUS SCENE

• **Stage 0: Dormant Quad (0 pts):** Muted grounds, bare paths, dry central quad with zero green infrastructure.

• **Stage 1: Native Shade Trees (30 pts):** Lush lawn grass, canopy shade trees, and wildflower blooms appear.

• **Stage 2: Rooftop Solar Array (40 pts):** Photovoltaic solar panels added to the academic hall roof.

• **Stage 3: Smart Compost Hubs (50 pts):** Color-coded organic compost pods and recycling units along campus walks.

• **Stage 4: Pure Water Cistern (60 pts):** Chilled water refill station and fountain basin complete the quad.

📷 [INSERT SCREENSHOT: Home Screen & Campus Scene]

Capture directly from the Android phone running EcoQuest in Expo Go

- Campus Restoration Scene at Stage 1 or Stage 4
- Level badge and animated progress bar
- Restoration Project Card showing cost and build action
- Lifetime Points, Spendable Fund, and Quests count

Sustainability Quests & Two-Tier Points Economy

💰 THE TWO-TIER ECONOMIC MODEL

• **8 Quests across 5 Domains:** Waste (ch-1, ch-5, ch-8), Energy (ch-2, ch-7), Transport (ch-3), Food (ch-4), Water (ch-6).

• **Lifetime Points (Civic Score):** Max 210 points. Never decreases when upgrades are purchased; drives Player Level, Badges, and Leaderboard.

• **Spendable Points (Restoration Fund):** Dedicated currency for funding campus upgrades. Deducted upon upgrade purchase.

• **Surplus Equation Balance:** 210 Lifetime Points Earned − 180 Upgrade Costs = **30 Spendable Surplus** remaining upon full quad completion.

📷 [INSERT SCREENSHOT: Quests Catalog Screen]

Capture directly from the Android phone running EcoQuest in Expo Go

- Category filter pills (All, Energy, Waste, etc.)
- Interactive cards with impact tags and point badges
- One-tap 'Mark as Done' buttons with instant status change
- Completed quests show disabled '✓ Completed' badges

Player Progression: Levels, Badges & Leaderboard

LEVELS & BADGE PROGRESSION

• **4 Player Level Tiers:** L1: Eco Novice (0–49) → L2: Campus Steward (50–99) → L3: Green Pioneer (100–169) → L4: Restoration Champion (170–210).

• **4 Score Milestone Badges:** First Step (1 quest), Century Club (100 pts), Earth Guardian (160 pts), Campus Hero (200 pts).

• **Critical Invariant:** Earth Guardian badge unlocks at 160 pts, verified distinct from Level 4 at 170 pts.

• **4 Domain Mastery Badges:** Zero Waste Hero (2 waste), Energy Guardian (both energy), Clean Cruiser (ch-3), Water Steward (ch-6).

• **Dynamic Leaderboard:** User starts at #8 and climbs in real time to Rank #4 alongside 7 benchmark peers.

[INSERT SCREENSHOT: Rank or Badges Screen]

Capture directly from the Android phone running EcoQuest in Expo Go

- Dynamic Leaderboard with demonstrator climbing to Rank #4
- Top-3 Podium showing Maya (1st), Aarav (2nd), Priya (3rd)
- Badges Showcase grid with unlocked vs. locked trophies
- Real-time category progress bars (e.g., 1/2 Waste quests)

Offline Persistence & Two-Step Demo Reset

💾 OFFLINE RESILIENCE & LIFECYCLE

• **AsyncStorage Persistence:** Game state persists under 'ecoquest.gameState.v1', surviving app reloads and Expo Go restarts.

• **Safe Hydration Splash:** Themed loading screen during initial storage read prevents false zero-point flashes.

• **Sequential Write Serialization:** Asynchronous mutex queue guarantees rapid quest completions write in strict order.

• **Schema Validation & Recovery:** Automatically purges corrupted JSON, strips unknown IDs, and repairs non-sequential upgrades.

• **Two-Step Demo Reset Dialog:** Profile tab includes confirmed reset (Cancel vs. Reset Everything) returning app to Stage 0 instantly.

📷 [INSERT SCREENSHOT: Profile & Reset Dialog]

Capture directly from the Android phone running EcoQuest in Expo Go

- Demonstrator account profile card and statistics
- Expo SDK 57 runtime and offline storage indicators
- Native Alert.alert confirmation modal popup
- 'Cancel' and 'Reset Everything' demonstration buttons

Verification, Limitations & Future Roadmap

🧪 TEST EVIDENCE

• **83 / 83 Assertions Passed:** `tests/test_integration.mjs` executed via Node.js with 100% success rate.

• **Invariants Verified:** 210 max pts, 180 upgrade costs, 30 surplus pts, duplicate blocking.

• **Bundler Compilation:** Metro bundler compiled 763 modules with 0 syntax warnings (HTTP 200 OK).

• **Android Layout:** Safe-area tested for notch and 48dp bottom nav clearance.

⚠️ PROJECT LIMITATIONS

• **Self-Reported Actions:** No physical camera AI or IoT sensor verification of actual student habits.

• **Simulated Peers:** 7 competitors are scripted benchmarks rather than live networked students.

• **Client-Side Storage:** Progress stored in AsyncStorage; no multi-device cloud account sync.

• **Pedagogical Estimates:** Impact labels are educational estimates, not certified carbon offsets.

🚀 FUTURE ROADMAP

• **Cloud Backend:** Node.js / PostgreSQL backend with university SSO authentication.

• **QR-Code Verification:** Physical QR codes at campus recycling pods and bicycle racks to verify actions.

• **Inter-Hostel Battles:** Live multiplayer competition between departments and student residences.

• **Admin Dashboard:** Campus sustainability committee web portal for anonymous resource tracking.

PROJECT CONCLUSION & DEMONSTRATION

Summary of Outcomes & Live Mobile Demo

KEY PROJECT ACHIEVEMENTS

• **Full-Featured Gamification Prototype:** Delivered 5 tabs, 8 quests, 4 upgrades, 4 levels, 8 badges, and offline persistence.

• **Behavioral Reinforcement:** Transforms passive guidelines into an engaging, visual campus restoration experience.

• **Robust Software Engineering:** Verified with 83 automated tests (100% pass rate) and clean Expo SDK 57 bundling.

• **Zero Server Cost:** Self-contained offline architecture ideal for campus community deployment.

LIVE ANDROID DEMO SEQUENCE

1. Baseline State (Stage 0): Observe dry, dormant campus quad with 0 points and locked upgrades.

2. Quest Completion: Switch to Quests tab; complete Waste & Energy quests to earn points.

3. Visual Restoration: Return to Home tab; fund Native Trees & Rooftop Solar Array (Stages 1 & 2).

4. Progression & Ranks: Inspect Rank tab (user climbs up) and Badges tab (trophies unlock).

5. Demo Reset: Profile tab → confirmed reset restores Stage 0 in milliseconds.